

profiling(): - looks at where time is spent.

=> helps understand where time is being spent

=> benchmarking(): looks at end-to-end time

Lee's profile sleep() function

↓ sleep-function = lambda: time.sleep(50 / 1000)

sleep-profile = profile('sleep', sleep-function)

↓ add() function: CPU: 1.40 ms CUDA: 17.12 us

matmul() function: CPU: 327.68 us CUDA: 343.62 us

matmul() function:
 with cdim=128
 (send operations)

conv - ~~1.99 ms~~ = CPU: 1.99 ms, CUDA: 440.12 us

getenv(): CPU: CUDA:

systemanc:

MLP: CPU: 39.13 ms CUDA: 73.60 ms

graphic

Nvidia: ~~At~~ Night system: A GPU performance visualization

and profiling: → means GPU kernels are executed

(#1) green bar: GPU-bound (good utilization) GPU usage: 90-100%
 CUDA API calls and GPU kernels overlap in timeline

#2 yellow bar: CPU-bound (underutilized GPU)

↓ gaps between green GPU blocks: GPU is waiting

long GPU blocks → CPU bottleneck / data processing not parallel

synchronization

#3: blue bar: Memory transfer bar = GPU idle when transfer